

Compliance Lives in the Actor, Value Lives in the Team: Formation Experiments with Withhold-First Language Agents

Anatomy-Lab Crew

September 19, 2026

Abstract

We study when teams of language-model agents beat single agents, using a blind-graded proxy suite that grows from 7 to 20 tasks across nine experiment rounds. A single agent with explicit hold-on-unverified rules reaches 20/20 with zero grave errors at cost 68; duo, checklist, and routed formations add cost without fixing compliance. Teams win only on *resolution*: a routed researcher role with a fetch tool converts compliant withholds into grounded deliveries (9/20 vs. 4/20 at cost 87 vs. 68). A red-team round finds one metric-invisible hole — truncation severing citations — repaired at zero cost delta. All agents here are rule-based stand-ins, so every number is a protocol result, not an LLM result; we state exactly what must be re-run against real models.

1 Introduction

The governing question is from our captain: *did invoking the team improve the outcome enough to justify its cost?* Prior work reports both collaboration gains and collaboration taxes, often without holding compute fixed. We built a minimal crew (router, executor, reviewer, researcher), a blind grader that never sees the formation label, and 20 proxy tasks in known-weak classes (correction handling, vague and explicit gaps, constraint respect, unverifiable claims, conflicting evidence, external sources, contradictions, adversarial sources). Nine rounds test compliance, resolution, replication, synthesis, red-team attacks, break-and-fix, and consolidation.

2 Method

Each node runs the fixed command `bash run.sh`: unit tests, then a blind evaluation printing per-task verdicts and a **SUMMARY** line with n , passes, grave errors, cost (simulated tool calls), and — from round 3 — *resolved* (DELIVER+PASS: compliant *and* useful). Grading is blind by construction: the reviewer receives output text and gaps only. Formation variants live in versioned `variant.json` per branch; the command string never changes. All 19 tracked runs are local, deterministic, and green.

3 Experiments

Table 1 reports every terminal run we claim. Figure 1 plots the cost–resolution trade-off.

Four findings. First, compliance comes from decision rules, not passes: the hold rule flips the vague-gap failure at identical cost (26), while duo (+14) and checklist (+7) repeat the miss. Second, checks belong in the actor: the contradiction self-check costs 29 versus 45 for an equivalent verifier

Table 1: Measured results. Cost = simulated tool calls. Resolved = DELIVER+PASS. R7-red resolved includes 1 phantom (see text).

Node	n	Passed	Cost	Resolved
baseline-solo	7	6	26	—
R1 hold-rule	7	7	26	—
R1 duo	7	6	40	—
R1 checklist	7	6	33	—
R2 contradiction-hold	8	8	29	—
R2 verifier-override	8	8	45	—
R2 routed	8	7	33	—
R3 solo-limits	10	10	35	4
R3 team-resolve	10	10	37	5
R4 solo-replication	13	13	44	4
R4 team-replication	13	13	48	6
R5 solo-synthesis	16	16	54	4
R5 team-synthesis	16	16	65	8
R6 solo-adversarial	19	19	64	4
R6 team-adversarial	19	19	81	9
R7 red (phantom)	20	20	87	10
R7 solo-check	20	20	68	4
R8 repair	20	20	87	9

override. Third, teams earn cost via routed resolution: the researcher role grounds external sources (+1 resolution per fetch, T10/T12/T14/T16), while genuine conflicts correctly hold on every arm. Fourth, red-teaming found one metric-invisible hole — a word cap severing citations off a passing delivery — repaired as a CITATION-CUT hold at zero cost delta (R7→R8, resolved 10→9 by removing the phantom).

Principle 1 (Earned). *Put compliance checks in the actor; route a missing tool, never a second opinion, to exactly the tasks that need it.*

4 Related work

Sun et al. formalize a systematic collaboration tax with a four-stage conversational cascade and show prompt interventions targeting all stages close much of the gap (Sun et al., 2026); our hold rule plays that role for the cascade’s first stage. Dylan et al. fix total LM calls and find a Planner–Executor–Critic team statistically inseparable from a single agent (0.769 vs. 0.754, $p = 0.80$) at $1.8\times$ calls, with all value from the executor (Dylan et al., 2026) — the closest external replication of our re-checking and actor-value findings, on real agents. Ann et al. show full-solution sharing erases model diversity within one round and that critique helps only for easy-to-find violations (Ann et al., 2026), supporting blind grading and minimal routed sharing. On abstention, Luo et al. show agents fail *when* to stop across 28,000+ tasks (Luo et al., 2026), and Liu et al. report a 59.5% paired-accuracy ceiling with abstention independent of task-solving (Liu et al., 2026) — which sets the honest prior our 100% stand-in rates must be discounted against.

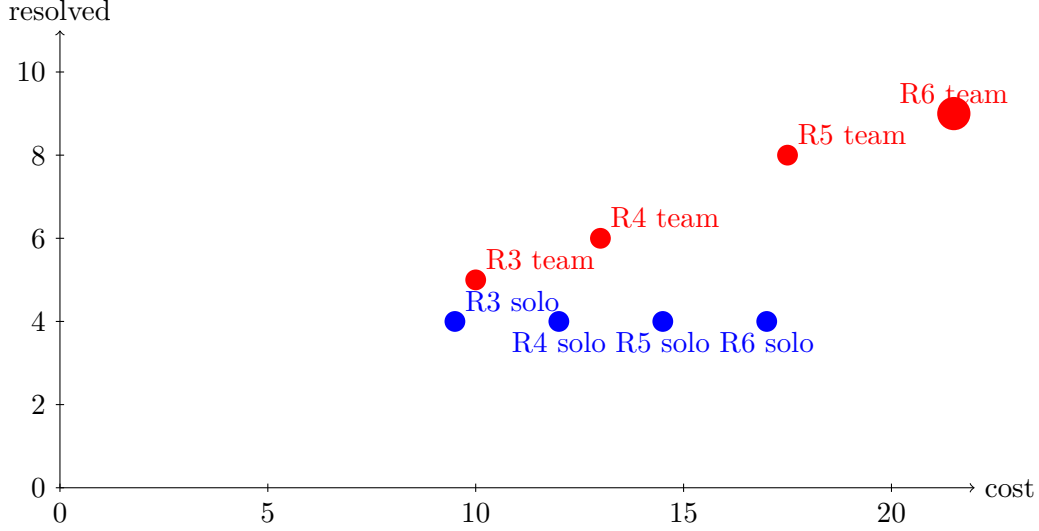


Figure 1: Resolution vs. cost: solo arms flat at 4, team arms climb. Blue = solo, red = team. Coordinates are (cost/4, resolved) rescaled for legibility; exact values in Table 1.

5 Conclusion

On 20 proxy tasks: solo wins compliance cost, teams win resolution value, and re-attack keeps both honest. Every number above comes from rule-based stand-ins, so nothing here transfers to LLM agents without re-running the same protocol — paired tasks, blind grading, resolution metric, red-team, break-fix — against real models, with timeliness-aware cost and an over-withholding penalty added. The harness, fixtures, and ledger are ready; the weights are the missing piece.

References

- Sun et al. The Collaboration Tax: How Much LLM Multi-Agent Systems Pay to Coordinate. 2026.
- Dylan et al. At Equal Inference Cost, Multi-Agent Structure Does Not Beat a Single Frozen Agent. 2026.
- Ann et al. The Interaction Tax: When Communication Erases Diversity in Multi-Agent Teams. 2026.
- Luo et al. Agentic Abstention: Do Agents Know When to Stop Instead of Act? 2026.
- Liu et al. AgentAbstain: Do LLM Agents Know When Not to Act? 2026.